

MARION ISLAND, South Africa

WMO: 689940

Lat: 46.877S

Lon: 37.859E

Elev: 79

StdP: 14.65

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	30.2	31.6	19.6	14.8	35.1	21.8	16.4	35.0	49.2	39.0	45.2	43.5	16.2	240	0.991

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.3	56.9	53.0	54.9	51.8	53.5	51.0	54.4	55.6	52.9	54.1	51.7	52.8	20.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.8	61.7	54.8	52.3	58.4	53.3	51.0	55.7	52.2	22.9	55.8	22.0	54.3	21.3	52.9	62.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.6	39.8	35.4	27.0	63.1	1.6	3.1	25.9	65.4	25.0	67.2	24.1	69.0	23.0	71.2		
(5)			25.1	58.0	1.4	2.1	24.1	59.5	23.3	60.7	22.5	61.9	21.4	63.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	46.4	47.0	45.9	44.3	42.4	40.6	40.0	39.6	39.5	41.0	42.5	44.7		
	DBStd	4.53	3.37	3.37	3.44	3.92	4.00	3.92	3.98	3.82	3.87	3.55	3.35	3.62		
	HDD50	2663	121	94	134	176	235	282	309	323	317	279	225	168		
	HDD65	8099	577	503	592	622	699	732	774	788	766	743	674	629		
	CDD50	38	9	10	7	5	1	0	0	0	0	1	1	4		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	17.6	17.0	16.1	16.7	16.8	17.6	17.8	19.1	18.6	19.0	17.8	17.1	17.5		
	PrecAvg	87.4	7.6	7.0	7.4	7.5	7.8	7.2	6.9	6.6	6.9	6.3	6.6	7.3		
(15) Precipitation	PrecMax	116.1	13.8	13.5	15.8	14.3	16.9	11.2	12.2	9.9	14.1	11.5	11.5	12.9		
	PrecMin	56.3	3.3	2.9	0.6	1.8	2.4	3.6	2.9	3.4	3.1	3.0	2.7	0.4		
	PrecStd	14.7	2.4	2.4	2.8	2.5	2.7	1.9	1.8	1.9	2.0	1.9	1.9	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	59.7	57.7	58.1	55.2	51.7	51.8	49.9	51.3	53.5	55.1	57.8		
		MCWB	54.2	52.8	53.9	53.6	51.5	48.5	48.3	48.1	48.4	49.9	51.9	52.9		
	2%	DB	55.8	55.9	54.8	54.2	52.0	49.6	48.8	47.9	48.3	50.1	51.5	54.0		
		MCWB	52.9	52.0	52.1	51.8	50.1	47.8	47.1	46.5	46.1	47.7	48.6	50.4		
	5%	DB	53.8	54.0	53.0	52.2	50.2	48.2	47.5	46.8	46.6	48.3	49.5	52.2		
		MCWB	51.2	51.3	51.0	50.5	48.7	47.0	46.1	45.6	45.1	46.5	46.8	49.3		
	10%	DB	52.0	52.5	51.6	50.4	48.8	46.9	46.2	45.7	45.4	46.7	48.1	50.5		
		MCWB	49.7	50.3	49.6	48.9	47.4	45.6	44.9	44.5	43.9	44.9	45.8	48.3		
	0.4%	WB	56.6	55.8	55.2	54.7	53.2	50.3	49.4	48.9	49.7	50.4	53.2	54.7		
		MCDB	57.4	57.2	56.7	56.0	54.1	50.8	50.4	49.3	50.7	51.8	55.1	56.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.9	53.5	53.1	52.5	50.6	48.5	47.7	47.0	46.8	48.3	49.6	51.9		
		MCDB	54.9	54.5	54.1	53.6	51.4	49.1	48.4	47.5	47.6	49.4	50.7	53.1		
	5%	WB	52.0	52.1	51.6	50.8	49.0	47.3	46.5	45.9	45.4	46.8	47.7	50.3		
		MCDB	53.2	53.1	52.6	51.9	49.9	48.0	47.3	46.5	46.3	47.8	48.8	51.4		
	10%	WB	50.2	50.8	50.1	49.2	47.6	45.8	45.2	44.7	44.2	45.3	46.4	48.9		
		MCDB	51.4	51.9	51.2	50.3	48.6	46.7	46.1	45.6	45.2	46.4	47.7	50.2		
	5% DB	MDBR	8.2	8.3	7.7	7.4	7.2	6.8	6.9	7.0	7.1	7.5	7.9	7.9		
		MCDBR	11.5	11.0	9.8	9.6	9.9	8.6	8.2	8.1	8.7	9.5	9.8	10.2		
		MCWBR	10.3	9.9	9.8	9.0	10.3	9.0	8.6	8.9	8.8	8.8	8.9	8.9		
		MCWBR	10.9	10.0	9.7	9.3	9.6	8.3	8.2	7.9	8.4	8.9	9.1	9.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.6	10.1	10.2	9.2	10.4	9.3	9.0	9.1	9.2	9.1	9.2	8.9		
	taub		0.353	0.345	0.342	0.328	0.332	0.336	0.345	0.353	0.364	0.357	0.353	0.352		
		taud	2.489	2.525	2.523	2.539	2.407	2.309	2.307	2.355	2.347	2.420	2.458	2.467		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		306	298	282	260	225	203	213	243	268	291	304	308		
		Edh at Noon	35	32	29	23	21	21	23	27	33	35	36	36		
	RadAvg		1661	1410	1005	610	383	268	315	528	851	1208	1531	1743		
(45) All-Sky Solar Radiation	RadStd		129	82	51	42	28	17	17	36	49	91	107	148		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			-0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page